

JOHNS HOPKINS UNIVERSITY PRESENTS

hyper_object

An Immersive Neuro-Theater Experience

Created by Prof. Graham Sack



Hopkins Bloomberg Center Theater

June 4, 2026



About the Performance

hyper_object is a live theater and interactive media experience that uses wearable neurotechnology to tell the story of a sentient planet that can perceive the thoughts of humankind. Created by an interdisciplinary team of artists, scholars, and engineers centered at Johns Hopkins University, the project blends live theater, AI, EEG-based neurotechnology, responsive video, and generative sound to explore the shifting boundaries between human, artificial, and planetary intelligence.

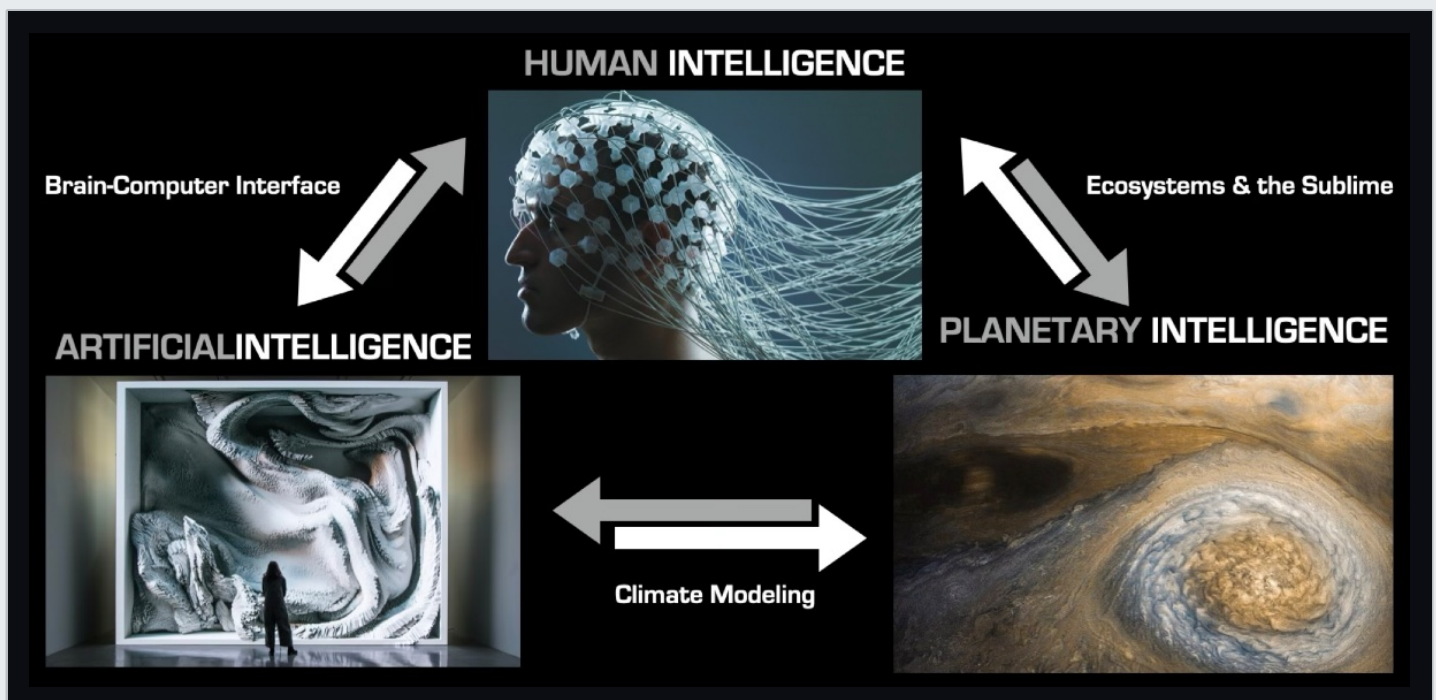
Synopsis

An astrobiologist arrives on a space station orbiting a mysterious exoplanet known as the hyperobject. As the crew's scientific mission unravels into confusion and paranoia, the planet begins to evoke memories, visions, and unconscious fears. The scientists must confront the possibility that the intelligence they are studying may understand them more deeply than they understand themselves.

Director's Statement

Graham Sack (Creator, Director, & Writer)

At the heart of hyper_object is a question: what happens when artificial intelligence seems to know us better than we know ourselves, while planetary systems reveal the limits of human prediction and control? By using live brainwave data to shape sound, image, and language in real time, the performance turns that question into a theatrical feedback loop among actors, audience, technology, and environment. The theater becomes a listening system, responsive not only to speech and movement, but to the invisible rhythms of the brain.



How EEG Data Shapes the Performance

Tonight's performance uses live EEG data from seven participants: one performer wearing a 32-channel Enobio headset, and six audience members in 4-channel Muse headsets. Their brainwave data is analyzed in real time and translated into five frequency bands.



01 Planetary Visualization

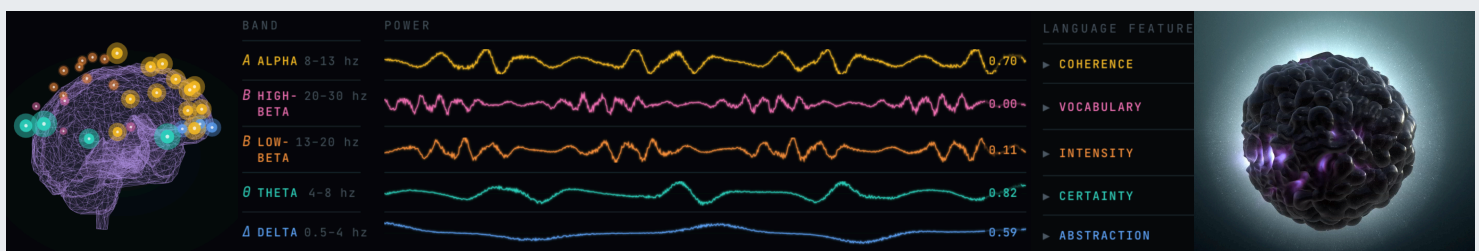
Audience and performer brainwaves drive the planetary entity projected onstage — its color, motion, displacement, and rotational velocity.

02 Sound Score

Audience EEG influences the score, changing the volume, speed, and presence of individual musical stems.

03 Responsive Dialogue

The performer's EEG helps drive an AI language system that generates the dialogue of Eva in real time.



A NOTE ON THE HEADSETS

The EEG headsets do not read private thoughts or memories. They measure broad patterns of electrical activity at the scalp and use those signals as artistic inputs. In the fiction of the piece, the hyperobject can perceive human thought; in the system of the performance, EEG becomes a way of making internal states visible, audible, and shared. **No data from tonight's performance is recorded or stored.**

CAST

David Skeist*

Kantor

T Ryder Smith*

Reinhardt

Catherine Gowl*

Eva

Jess Barbagallo*

Shaw

* Actor or Stage Manager appears through the courtesy of Actors' Equity Association, the Union of Professional Actors and Stage Managers in the United States

CREATIVE & PRODUCTION

Graham Sack

Creator, Writer, & Director

Quentin Little

Producer

Simon Harding

Production & Video Design

Brooke Bell*

Stage Manager

Matt McCorkle

Composer & Sound Design

Georgios Cherouvim

Planetary Developer

Suzanne Dikker

EEG Design

Franck Porteous

EEG Programming

Conor James

Developer & Camera Operator

Lee Whitman

Developer

Colin K. Bills

Lighting Design

Karen Boyer

Costume Design

Mekala Sridhar

Assoc. Producer

Sabrina Hyman

Casting Director

Sara Ritmiller

Hair & Makeup

Yannick Godts

Electrician

NEXUS RESEARCH TEAM

Graham Sack

Berman Institute of Bioethics
Film & Media Graduate Program

Debra Mathews (PI)

Berman Institute of Bioethics

Sig Libowitz, J.D.

Film & Media Graduate Program

Lauren Arora Hutchinson

Dracopoulos-Bloomberg iDeas Lab

Tom Lippincott

Center for Digital Humanities

Susan Magsamen

International Arts + Mind Lab

Nathan Crone, M.D.

School of Medicine

Bernadette Wegenstein

Center for Advanced Media Studies

Griffin Milsap

Whiting School of Engineering

GRANT & INSTITUTIONAL CONTEXT

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